

CREATE TABLE:-**PATIENTS TABLE:-**

CREATE TABLE Patients (

patient_id INT PRIMARY KEY,
name VARCHAR(100) NOT NULL,
dob DATE,
gender VARCHAR(10),
phone_number VARCHAR(15),
email VARCHAR(100),
address VARCHAR(200),
registration_date DATE

);

INSERT INTO Patients VALUES

(1,'Aarav Patel','2001-05-12','Male','9876543210','aarav@gmail.com','Ahmedabad','2026-01-05'),
(2,'Riya Shah','1999-08-20','Female','9876543211','riya@gmail.com','Surat','2026-01-10'),
(3,'Dev Mehta','1995-02-15','Male','9876543212','dev@gmail.com','Vadodara','2026-01-15'),
(4,'Anaya Desai','2003-11-10','Female','9876543213','anaya@gmail.com','Ahmedabad','2026-02-01'),
(5,'Vivaan Joshi','1988-06-25','Male','9876543214','vivaan@gmail.com','Rajkot','2026-02-05'),
(6,'Isha Patel','2000-03-18','Female','9876543215','isha@gmail.com','Ahmedabad','2026-02-12'),
(7,'Krish Shah','1992-09-30','Male','9876543216','krish@gmail.com','Surat','2026-02-20'),
(8,'Meera Trivedi','1985-12-05','Female','9876543217','meera@gmail.com','Vadodara','2026-03-01'),
(9,'Yash Thakkar','1998-07-14','Male','9876543218','yash@gmail.com','Ahmedabad','2026-03-10'),
(10,'Diya Patel','2004-04-22','Female','9876543219','diya@gmail.com','Rajkot','2026-03-15'),
(11,'Arjun Shah','1990-10-11','Male','9876543220','arjun@gmail.com','Ahmedabad','2026-04-01'),
(12,'Kavya Mehta','1997-01-28','Female','9876543221','kavya@gmail.com','Surat','2026-04-05'),
(13,'Manav Desai','1983-05-19','Male','9876543222','manav@gmail.com','Vadodara','2026-04-12'),
(14,'Pooja Joshi','1994-09-09','Female','9876543223','pooja@gmail.com','Ahmedabad','2026-05-01'),
(15,'Dhruv Patel','2002-02-27','Male','9876543224','dhruv@gmail.com','Surat','2026-05-10');

Show query box

15 rows inserted (Query took 0.0084 seconds)

```
INSERT INTO Patients VALUES (1,'Aarav Patel','2001-05-12','Male','9876543210','aarav@gmail.com','Ahmedabad','2026-01-05'), (2,'Riya Shah','1999-08-20','Female','9876543211','riya@gmail.com','Surat','2026-01-10'), (3,'Dev Mehta','1995-02-15','Male','9876543212','dev@gmail.com','Vadodara','2026-01-15'), (4,'Anaya Desai','2003-11-10','Female','9876543213','anaya@gmail.com','Ahmedabad','2026-02-01'), (5,'Vivaan Joshi','1988-06-25','Male','9876543214','vivaan@gmail.com','Rajkot','2026-02-05'), (6,'Isha Patel','2000-03-18','Female','9876543215','isha@gmail.com','Ahmedabad','2026-02-12'), (7,'Krish Shah','1992-09-30','Male','9876543216','krish@gmail.com','Surat','2026-02-20'), (8,'Meera Trivedi','1985-12-05','Female','9876543217','meera@gmail.com','Vadodara','2026-03-01'), (9,'Yash Thakkar','1998-07-14','Male','9876543218','yash@gmail.com','Ahmedabad','2026-03-10'), (10,'Diya Patel','2004-04-22','Female','9876543219','diya@gmail.com','Rajkot','2026-03-15'), (11,'Arjun Shah','1990-10-11','Male','9876543220','arjun@gmail.com','Ahmedabad','2026-04-01'), (12,'Kavya Mehta','1997-01-28','Female','9876543221','kavya@gmail.com','Surat','2026-04-05'), (13,'Manav Desai','1983-05-19','Male','9876543222','manav@gmail.com','Vadodara','2026-04-12'), (14,'Pooja Joshi','1994-09-09','Female','9876543223','pooja@gmail.com','Ahmedabad','2026-05-01'), (15,'Dhruv Patel','2002-02-27','Male','9876543224','dhruv@gmail.com','Surat','2026-05-10');
```

EXAM

DOCTOR TABLE:-

```
CREATE TABLE Doctors (  
  doctor_id INT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  specialization VARCHAR(100),  
  phone_number VARCHAR(15),  
  email VARCHAR(100),  
  available_days VARCHAR(100),  
  consultation_fee DECIMAL(10,2)  
);  
INSERT INTO Doctors VALUES  
(1,'Dr. Raj Sharma','Cardiology','9000000001','raj@hospital.com','Monday, Wednesday,  
Friday',1500),  
(2,'Dr. Neha Patel','Dermatology','9000000002','neha@hospital.com','Tuesday, Thursday',1000),  
(3,'Dr. Amit Shah','Neurology','9000000003','amit@hospital.com','Monday, Thursday',1800),  
(4,'Dr. Priya Mehta','Pediatrics','9000000004','priya@hospital.com','Monday, Wednesday',1200),  
(5,'Dr. Karan Desai','Orthopedics','9000000005','karan@hospital.com','Tuesday, Friday',1600),  
(6,'Dr. Rina Joshi','General Medicine','9000000006','rina@hospital.com','Monday, Tuesday,  
Friday',800),  
(7,'Dr. Vivek Trivedi','Cardiology','9000000007','vivek@hospital.com','Wednesday,  
Saturday',1700),  
(8,'Dr. Simran Shah','Dermatology','9000000008','simran@hospital.com','Thursday,  
Saturday',1100);
```



The screenshot shows a database query execution result. At the top, it says "8 rows inserted (Query took 0.0006 seconds)". Below this, the SQL statement is displayed, which is the same CREATE TABLE and INSERT INTO statement as shown in the text above. The bottom of the screenshot shows a status bar with "[08 rows] [08] [Create Table]".

EXAM

DEPARTMENT TABLE:-

```
CREATE TABLE Departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(100) NOT NULL  
);
```

```
INSERT INTO Departments VALUES  
(1,'Cardiology'),  
(2,'Dermatology'),  
(3,'Neurology'),  
(4,'Pediatrics'),  
(5,'Orthopedics'),  
(6,'General Medicine');
```



APPOINTMENT TABLE:-

```
CREATE TABLE Appointments (  
    appointment_id INT PRIMARY KEY,  
    patient_id INT,  
    doctor_id INT,  
    appointment_date DATE,  
    status VARCHAR(20),  
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)  
);
```

```
INSERT INTO Appointments VALUES  
(1,1,1,'2026-06-01','Completed'),  
(2,2,2,'2026-06-03','Scheduled'),  
(3,3,3,'2026-06-05','Completed'),  
(4,4,4,'2026-06-07','Completed'),  
(5,5,5,'2026-06-10','Cancelled'),  
(6,6,6,'2026-06-12','Completed'),  
(7,7,1,'2026-06-15','Scheduled'),  
(8,8,7,'2026-06-18','Completed'),  
(9,9,3,'2026-06-20','Completed'),  
(10,10,4,'2026-06-22','Scheduled'),  
(11,11,5,'2026-06-25','Completed'),  
(12,12,2,'2026-06-27','Cancelled'),  
(13,13,6,'2026-07-01','Completed');
```

EXAM

(14,14,8,'2026-07-03','Scheduled'),
(15,15,1,'2026-07-05','Completed');

```
✓ 15 rows inserted (Query took 0.0025 seconds)

INSERT INTO Appointment VALUES (1,1,1,'2026-06-01','Completed'), (1,2,1,'2026-06-01','Scheduled'), (1,3,1,'2026-06-01','Completed'), (4,4,4,'2026-06-07','Completed'), (4,5,4,'2026-06-07','Completed'), (4,6,4,'2026-06-07','Completed'), (4,7,4,'2026-06-07','Completed'), (4,8,4,'2026-06-07','Completed'), (4,9,4,'2026-06-07','Completed'), (4,10,4,'2026-06-07','Completed'), (4,11,4,'2026-06-07','Completed'), (4,12,4,'2026-06-07','Completed'), (4,13,4,'2026-06-07','Completed'), (4,14,4,'2026-06-07','Completed'), (4,15,4,'2026-06-07','Completed'), (10,10,4,'2026-06-22','Scheduled'), (11,11,6,'2026-07-01','Completed'), (12,12,6,'2026-07-01','Completed'), (13,13,6,'2026-07-01','Completed'), (14,14,8,'2026-07-03','Scheduled'), (15,15,1,'2026-07-05','Completed');

✓ 15 rows inserted (Query took 0.0025 seconds)
```

MEDICAL RECORD TABLE:-

```
CREATE TABLE Medical_Records (
    record_id INT PRIMARY KEY,
    patient_id INT,
    doctor_id INT,
    diagnosis VARCHAR(200),
    prescription VARCHAR(300),
    treatment_date DATE,
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)
);
```

```
INSERT INTO Medical_Records VALUES
(1,1,1,'High Blood Pressure','BP Medicine','2026-06-01'),
(2,2,2,'Skin Allergy','Antihistamine Cream','2026-06-03'),
(3,3,3,'Migraine','Pain Relief Tablets','2026-06-05'),
(4,4,4,'Fever','Paracetamol','2026-06-07'),
(5,6,6,'Viral Infection','Antibiotics','2026-06-12'),
(6,7,1,'Chest Pain','ECG and Heart Medicine','2026-06-15'),
(7,8,7,'Heart Problem','Heart Medicine','2026-06-18'),
(8,9,3,'Migraine','Migraine Tablets','2026-06-20'),
(9,10,4,'Cold and Fever','Syrup and Tablets','2026-06-22'),
(10,11,5,'Knee Pain','Painkiller and Exercise','2026-06-25'),
(11,13,6,'Diabetes','Diabetes Medicine','2026-07-01'),
(12,15,1,'Cholesterol','Cholesterol Medicine','2026-07-05');
```

```
✓ 12 rows inserted (Query took 0.0006 seconds)

INSERT INTO Medical_Records VALUES (1,1,1,'High Blood Pressure','BP Medicine','2026-06-01'), (2,2,2,'Skin Allergy','Antihistamine Cream','2026-06-03'), (3,3,3,'Migraine','Pain Relief Tablets','2026-06-05'), (4,4,4,'Fever','Paracetamol','2026-06-07'), (5,6,6,'Viral Infection','Antibiotics','2026-06-12'), (6,7,1,'Chest Pain','ECG and Heart Medicine','2026-06-15'), (7,8,7,'Heart Problem','Heart Medicine','2026-06-18'), (8,9,3,'Migraine','Migraine Tablets','2026-06-20'), (9,10,4,'Cold and Fever','Syrup and Tablets','2026-06-22'), (10,11,5,'Knee Pain','Painkiller and Exercise','2026-06-25'), (11,13,6,'Diabetes','Diabetes Medicine','2026-07-01'), (12,15,1,'Cholesterol','Cholesterol Medicine','2026-07-05');

✓ 12 rows inserted (Query took 0.0006 seconds)
```

EXAM

BILLING TABLE:-

```
CREATE TABLE Billing (  
    invoice_id INT PRIMARY KEY,  
    patient_id INT,  
    appointment_id INT,  
    amount DECIMAL(10,2),  
    payment_status VARCHAR(20),  
    payment_date DATE,  
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
    FOREIGN KEY (appointment_id) REFERENCES Appointments(appointment_id)  
);
```

```
INSERT INTO Billing VALUES  
(1,1,1,2500,'Paid','2026-06-01'),  
(2,2,2,1500,'Pending','2026-06-03'),  
(3,3,3,3000,'Paid','2026-06-05'),  
(4,4,4,1800,'Paid','2026-06-07'),  
(5,5,5,1600,'Cancelled','2026-06-10'),  
(6,6,6,1200,'Paid','2026-06-12'),  
(7,7,7,2500,'Pending','2026-06-15'),  
(8,8,8,2800,'Paid','2026-06-18'),  
(9,9,9,3000,'Paid','2026-06-20'),  
(10,10,10,1800,'Pending','2026-06-22'),  
(11,11,11,2600,'Paid','2026-06-25'),  
(12,12,12,1500,'Cancelled','2026-06-27'),  
(13,13,13,1200,'Paid','2026-07-01'),  
(14,14,14,1600,'Pending','2026-07-03'),  
(15,15,15,2500,'Paid','2026-07-05');
```

15 rows inserted. (Query took 0.0008 seconds.)

```
INSERT INTO Billing VALUES (1,1,1,2500,'Paid','2026-06-01'), (2,2,2,1500,'Pending','2026-06-03'), (3,3,3,3000,'Paid','2026-06-05'), (4,4,4,1800,'Paid','2026-06-07'), (5,5,5,1600,'Cancelled','2026-06-10'),  
(6,6,6,1200,'Paid','2026-06-12'), (7,7,7,2500,'Pending','2026-06-15'), (8,8,8,2800,'Paid','2026-06-18'), (9,9,9,3000,'Paid','2026-06-20'), (10,10,10,1800,'Pending','2026-06-22'), (11,11,11,2600,'Paid','2026-06-25'),  
(12,12,12,1500,'Cancelled','2026-06-27'), (13,13,13,1200,'Paid','2026-07-01'), (14,14,14,1600,'Pending','2026-07-03'), (15,15,15,2500,'Paid','2026-07-05');
```

[Click here] [15.00] [Create PHP code]

EXAM

DOCTOR DEPARTMENT TABLE:-

```
CREATE TABLE Doctor_Departments (  
  doctor_id INT,  
  department_id INT,  
  PRIMARY KEY (doctor_id, department_id),  
  FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),  
  FOREIGN KEY (department_id) REFERENCES Departments(department_id)  
);
```

INSERT INTO Doctor_Departments VALUES

```
(1,1),  
(2,2),  
(3,3),  
(4,4),  
(5,5),  
(6,6),  
(7,1),  
(8,2);
```



8 rows inserted (Query took 0.0007 seconds)

```
INSERT INTO Doctor_Departments VALUES (1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (7,1), (8,2);
```

Click here to edit this code in PHP editor

1. CRUD OPERATIONS:-

- INSERT

INSERT INTO Patients

VALUES

```
(16,'Nisha Patel','2001-10-15','Female','9876543225',  
'nisha@gmail.com','Ahmedabad','2026-07-10');
```

[Show query box](#)

```
✓ 1 row inserted. (Query took 0.0004 seconds.)  
  
INSERT INTO Patients VALUES (16,'Nisha Patel','2001-10-15','Female','9876543225','nisha@gmail.com','Ahmedabad','2026-07-10');  
  
[[Ctrl+Enter]] [[Ctrl+Z]] [[Close PHP code]]
```

- UPDATE

UPDATE Patients

SET phone_number = '9999999999',

address = 'Gandhinagar'

WHERE patient_id = 16;

[Show query box](#)

```
✓ 1 row affected. (Query took 0.0006 seconds.)  
  
UPDATE Patients SET phone_number = '9999999999', address = 'Gandhinagar' WHERE patient_id = 16;  
  
[[Ctrl+Enter]] [[Ctrl+Z]] [[Close PHP code]]
```

- DELETE

DELETE FROM Patients

WHERE patient_id = 16;

```
✓ 1 row deleted. (Query took 0.0014 seconds.)  
  
DELETE FROM Patients WHERE patient_id = 16;  
  
[[Ctrl+Enter]] [[Ctrl+Z]] [[Close PHP code]]
```

2. SQL CLAUSES:-

- LIMIT
SELECT * FROM Patients
LIMIT 20;

Extra options

		patient_id	name	dob	gender	phone_number	email	address	registration_date
☐	Edit	1	Aarav Patel	2001-05-12	Male	9876543210	aarav@gmail.com	Ahmedabad	2026-01-05
☐	Edit	2	Riya Shah	1999-08-20	Female	9876543211	riya@gmail.com	Surat	2026-01-10
☐	Edit	3	Dev Mehta	1995-02-15	Male	9876543212	dev@gmail.com	Vadodara	2026-01-15
☐	Edit	4	Anaya Dasai	2003-11-10	Female	9876543213	anaya@gmail.com	Ahmedabad	2026-02-01
☐	Edit	5	Vivaan Joshi	1988-06-25	Male	9876543214	vivaan@gmail.com	Rajkot	2026-02-05
☐	Edit	6	Isha Patel	2000-03-18	Female	9876543215	isha@gmail.com	Ahmedabad	2026-02-12
☐	Edit	7	Krish Shah	1992-09-30	Male	9876543216	krish@gmail.com	Surat	2026-02-20
☐	Edit	8	Meera Trivedi	1985-12-05	Female	9876543217	meera@gmail.com	Vadodara	2026-03-01
☐	Edit	9	Yash Thakkar	1996-07-14	Male	9876543218	yash@gmail.com	Ahmedabad	2026-03-10
☐	Edit	10	Diya Patel	2004-04-22	Female	9876543219	diya@gmail.com	Rajkot	2026-03-15
☐	Edit	11	Arjun Shah	1990-10-11	Male	9876543220	arjun@gmail.com	Ahmedabad	2026-04-01
☐	Edit	12	Kavya Mehta	1997-01-28	Female	9876543221	kavya@gmail.com	Surat	2026-04-05
☐	Edit	13	Manav Desai	1983-05-19	Male	9876543222	manav@gmail.com	Vadodara	2026-04-12
☐	Edit	14	Pooja Joshi	1994-09-09	Female	9876543223	pooja@gmail.com	Ahmedabad	2026-05-01
☐	Edit	15	Dhruv Patel	2002-02-27	Male	9876543224	dhruv@gmail.com	Surat	2026-05-10

EXAM

- WHERE
SELECT * FROM Doctors
WHERE consultation_fee > 1500;

doctor_id	name	specialization	phone_number	email	available_days	consultation_fee
3	Dr. Amit Shah	Neurology	9000000003	amit@hospital.com	Monday, Thursday	1000.00
5	Dr. Karan Desai	Orthopedics	9000000005	karan@hospital.com	Tuesday, Friday	1500.00
7	Dr. Vivek Trivedi	Cardiology	9000000007	vivek@hospital.com	Wednesday, Saturday	1700.00

- HIGHEST PAYING
SELECT p.patient_id, p.name, SUM(b.amount) AS total_amount
FROM Patients p
JOIN Billing b ON p.patient_id = b.patient_id
GROUP BY p.patient_id, p.name
ORDER BY total_amount DESC
LIMIT 5;

patient_id	name	total_amount
9	Yash Thakkar	3000.00
3	Dev Mehta	3000.00
8	Meera Trivedi	2800.00
11	Arjun Shah	2600.00
15	Dhruv Patel	2500.00

3. SQL OPERATORS :-

- AND
SELECT *
FROM Appointments
WHERE status = 'Scheduled'
AND doctor_id = 1;

extra options

	appointment_id	patient_id	doctor_id	appointment_date	status
Edit Copy Delete	7	7	1	2026-06-15	Scheduled
☐ Check all With selected: Edit Copy Delete Export					

- OR
SELECT *
FROM Doctors
WHERE specialization = 'Cardiology'
OR specialization = 'Neurology';

	doctor_id	name	specialization	phone_number	email	available_days	consultation_fee
Edit Copy Delete	1	Dr. Raj Sharma	Cardiology	9000000001	raj@hospital.com	Monday, Wednesday, Friday	1500.00
Edit Copy Delete	3	Dr. Amit Shah	Neurology	9000000003	amit@hospital.com	Monday, Thursday	1000.00
Edit Copy Delete	7	Dr. Vivek Trivedi	Cardiology	9000000007	vivek@hospital.com	Wednesday, Saturday	1700.00

- NOT
SELECT *
FROM Appointments
WHERE NOT status = 'Cancelled';

EXAM

Extra options

		appointment_id	patient_id	doctor_id	appointment_date	status
☐	 Edit  Copy  Delete	1	1	1	2026-06-01	Completed
☐	 Edit  Copy  Delete	2	2	2	2026-06-03	Scheduled
☐	 Edit  Copy  Delete	3	3	3	2026-06-05	Completed
☐	 Edit  Copy  Delete	4	4	4	2026-06-07	Completed
☐	 Edit  Copy  Delete	6	6	6	2026-06-12	Completed
☐	 Edit  Copy  Delete	7	7	1	2026-06-15	Scheduled
☐	 Edit  Copy  Delete	8	8	7	2026-06-18	Completed
☐	 Edit  Copy  Delete	9	9	3	2026-06-20	Completed
☐	 Edit  Copy  Delete	10	10	4	2026-06-22	Scheduled
☐	 Edit  Copy  Delete	11	11	5	2026-06-25	Completed
☐	 Edit  Copy  Delete	13	13	6	2026-07-01	Completed
☐	 Edit  Copy  Delete	14	14	8	2026-07-03	Scheduled
☐	 Edit  Copy  Delete	15	15	1	2026-07-05	Completed



☐ Check all

With selected:

 Edit

 Copy

 Delete

 Export

4. SORTING & GROUPING DATA:-

- SELECT *
- FROM Patients
- ORDER BY name ASC;

Show query box

Showing rows 0 - 14 (15 total, Query took 0.0102 seconds.) [name: AARAV PATEL... - YASH THAKKAR...]

SELECT * FROM Patients ORDER BY name ASC;

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	patient_id	name	dob	gender	phone_number	email	address	registration_date
☐ Edit Copy Delete	1	Aarav Patel	2001-05-12	Male	9876543210	aarav@gmail.com	Ahmedabad	2026-01-05
☐ Edit Copy Delete	4	Anaya Desai	2003-11-10	Female	9876543213	anaya@gmail.com	Ahmedabad	2026-02-01
☐ Edit Copy Delete	11	Arjun Shah	1990-10-11	Male	9876543220	arjun@gmail.com	Ahmedabad	2026-04-01
☐ Edit Copy Delete	3	Dev Mehta	1995-02-15	Male	9876543212	dev@gmail.com	Vadodara	2026-01-15
☐ Edit Copy Delete	15	Dhruv Patel	2002-02-27	Male	9876543224	dhruv@gmail.com	Surat	2026-05-10
☐ Edit Copy Delete	10	Diya Patel	2004-04-22	Female	9876543219	diya@gmail.com	Rajkot	2026-03-15
☐ Edit Copy Delete	6	Isha Patel	2000-03-18	Female	9876543215	isha@gmail.com	Ahmedabad	2026-02-12
☐ Edit Copy Delete	12	Kavya Mehta	1997-01-28	Female	9876543221	kavya@gmail.com	Surat	2026-04-05
☐ Edit Copy Delete	7	Krish Shah	1992-09-30	Male	9876543216	krish@gmail.com	Surat	2026-02-20
☐ Edit Copy Delete	13	Manav Desai	1983-05-19	Male	9876543222	manav@gmail.com	Vadodara	2026-04-12
☐ Edit Copy Delete	8	Meera Trivedi	1985-12-05	Female	9876543217	meera@gmail.com	Vadodara	2026-03-01
☐ Edit Copy Delete	14	Pooja Joshi	1994-09-09	Female	9876543223	pooja@gmail.com	Ahmedabad	2026-05-01
☐ Edit Copy Delete	2	Riya Shah	1999-08-20	Female	9876543211	riya@gmail.com	Surat	2026-01-10
☐ Edit Copy Delete	5	Vivaan Joshi	1988-06-25	Male	9876543214	vivaan@gmail.com	Rajkot	2026-02-05
☐ Edit Copy Delete	9	Yash Thakkar	1996-07-14	Male	9876543218	yash@gmail.com	Ahmedabad	2026-03-10

☐ Check all With selected: Edit Copy Delete Export

EXAM

- SELECT *
FROM Doctors
ORDER BY consultation_fee DESC;

data options

	doctor_id	name	specialization	phone_number	email	available_days	consultation_fee
  	3	Dr. Amit Shah	Neurology	9000000003	amit@hospital.com	Monday, Thursday	1800.00
  	7	Dr. Vivek Trivedi	Cardiology	9000000007	vivek@hospital.com	Wednesday, Saturday	1700.00
  	5	Dr. Karan Desai	Orthopedics	9000000005	karan@hospital.com	Tuesday, Friday	1600.00
  	1	Dr. Raj Sharma	Cardiology	9000000001	raj@hospital.com	Monday, Wednesday, Friday	1500.00
  	4	Dr. Priya Mehta	Pediatrics	9000000004	priya@hospital.com	Monday, Wednesday	1200.00
  	8	Dr. Simran Shah	Dermatology	9000000008	simran@hospital.com	Thursday, Saturday	1100.00
  	2	Dr. Neha Patel	Dermatology	9000000002	neha@hospital.com	Tuesday, Thursday	1000.00
  	6	Dr. Rina Joshi	General Medicine	9000000006	rina@hospital.com	Monday, Tuesday, Friday	800.00

☐ Check all With selected:    

☐ Show all Number of rows: 25 Filter rows: Sort by key: None

Query results operations

- SELECT gender, COUNT(*) AS total_patients
FROM Patients
GROUP BY gender;

gender	total_patients
Female	7
Male	8

☐ Show all Number of rows:

5.AGGREGATE FUNCTION :-

- Total
SELECT SUM(amount) AS total_revenue
FROM Billing
WHERE payment_status = 'Paid';

Extra options

total_revenue

20600.00

☐

Show all

Number

- Maximum
SELECT MAX(amount) AS maximum_bill
FROM Billing;

maximum_bill

3000.00

☐

Show all

Number

- Minimum
SELECT MIN(amount) AS minimum_bill
FROM Billing;

minimum_bill

1200.00

EXAM

- Average
SELECT AVG(amount) AS average_bill
FROM Billing;

Extra options

average_bill

2073.333333

6. FOREIGN&PRIMARY KEY RELATION:-

- PATIENT HAVING APOINTMENT
SELECT p.patient_id, p.name, a.appointment_date
FROM Patients p
INNER JOIN Appointments a
ON p.patient_id = a.patient_id;

patient_id	name	appointment_date
1	Aarav Patel	2026-06-01
2	Riya Shah	2026-06-03
3	Dev Mehta	2026-06-05
4	Anaya Desai	2026-06-07
5	Vivaan Joshi	2026-06-10
6	Isha Patel	2026-06-12
7	Krish Shah	2026-06-15
8	Meera Trivedi	2026-06-18
9	Yash Thakkar	2026-06-20
10	Diya Patel	2026-06-22
11	Arjun Shah	2026-06-25
12	Kavya Mehta	2026-06-27
13	Manav Desai	2026-07-01
14	Pooja Joshi	2026-07-03
15	Dhruv Patel	2026-07-05

EXAM

- DOCTOR HAVING APPOINTMENT
SELECT d.doctor_id, d.name, a.appointment_date
FROM Doctors d
INNER JOIN Appointments a
ON d.doctor_id = a.doctor_id;

Extra options

doctor_id	name	appointment_date
1	Dr. Raj Sharma	2026-06-01
1	Dr. Raj Sharma	2026-06-15
1	Dr. Raj Sharma	2026-07-05
2	Dr. Neha Patel	2026-06-03
2	Dr. Neha Patel	2026-06-27
3	Dr. Amit Shah	2026-06-05
3	Dr. Amit Shah	2026-06-20
4	Dr. Priya Mehta	2026-06-07
4	Dr. Priya Mehta	2026-06-22
5	Dr. Karan Desai	2026-06-10
5	Dr. Karan Desai	2026-06-25
6	Dr. Rina Joshi	2026-06-12
6	Dr. Rina Joshi	2026-07-01
7	Dr. Vivek Trivedi	2026-06-18
8	Dr. Simran Shah	2026-07-03

7.JOINS:-

- INNER JOIN
 SELECT p.name AS patient_name,
 d.name AS doctor_name,
 a.appointment_date
 FROM Appointments a
 INNER JOIN Patients p
 ON a.patient_id = p.patient_id
 INNER JOIN Doctors d
 ON a.doctor_id = d.doctor_id;

☐ Show all | Number of rows: 25 ▼ | Filter

Extra options

patient_name	doctor_name	appointment_date
Aarav Patel	Dr. Raj Sharma	2026-06-01
Krish Shah	Dr. Raj Sharma	2026-06-15
Dhruv Patel	Dr. Raj Sharma	2026-07-05
Riya Shah	Dr. Neha Patel	2026-06-03
Kavya Mehta	Dr. Neha Patel	2026-06-27
Dev Mehta	Dr. Amit Shah	2026-06-05
Yash Thakkar	Dr. Amit Shah	2026-06-20
Anaya Desai	Dr. Priya Mehta	2026-06-07
Diya Patel	Dr. Priya Mehta	2026-06-22
Vivaan Joshi	Dr. Karan Desai	2026-06-10
Arjun Shah	Dr. Karan Desai	2026-06-25
Isha Patel	Dr. Rina Joshi	2026-06-12
Manav Desai	Dr. Rina Joshi	2026-07-01
Meera Trivedi	Dr. Vivek Trivedi	2026-06-18
Pooja Joshi	Dr. Simran Shah	2026-07-03

EXAM

- LEFT JOIN
SELECT p.name AS patient_name,
a.appointment_date
FROM Patients p
LEFT JOIN Appointments a
ON p.patient_id = a.patient_id;

Extra options

patient_name	appointment_date
Aarav Patel	2026-06-01
Riya Shah	2026-06-03
Dev Mehta	2026-06-05
Anaya Desai	2026-06-07
Aivaan Joshi	2026-06-10
Asha Patel	2026-06-12
Krish Shah	2026-06-15
Meera Trivedi	2026-06-18
Rash Thakkar	2026-06-20
Diya Patel	2026-06-22
Arjun Shah	2026-06-25
Kavya Mehta	2026-06-27
Manav Desai	2026-07-01
Pooja Joshi	2026-07-03
Ahruv Patel	2026-07-05

EXAM

- RIGHT JOIN

```
SELECT d.name AS doctor_name,  
       a.appointment_date  
FROM Appointments a  
RIGHT JOIN Doctors d  
ON a.doctor_id = d.doctor_id;
```

Extra options

doctor_name	appointment_date
-------------	------------------

Dr. Raj Sharma	2026-06-01
----------------	------------

Dr. Raj Sharma	2026-06-15
----------------	------------

Dr. Raj Sharma	2026-07-05
----------------	------------

Dr. Neha Patel	2026-06-03
----------------	------------

Dr. Neha Patel	2026-06-27
----------------	------------

Dr. Amit Shah	2026-06-05
---------------	------------

Dr. Amit Shah	2026-06-20
---------------	------------

Dr. Priya Mehta	2026-06-07
-----------------	------------

Dr. Priya Mehta	2026-06-22
-----------------	------------

Dr. Karan Desai	2026-06-10
-----------------	------------

Dr. Karan Desai	2026-06-25
-----------------	------------

Dr. Rina Joshi	2026-06-12
----------------	------------

Dr. Rina Joshi	2026-07-01
----------------	------------

Dr. Vivek Trivedi	2026-06-18
-------------------	------------

Dr. Simran Shah	2026-07-03
-----------------	------------

- FULL OUTER JOIN
SELECT p.name, a.appointment_date
FROM Patients p
LEFT JOIN Appointments a
ON p.patient_id = a.patient_id
UNION
SELECT p.name, a.appointment_date
FROM Patients p
RIGHT JOIN Appointments a
ON p.patient_id = a.patient_id;

Extra options

name	appointment_date
Aarav Patel	2026-06-01
Riya Shah	2026-06-03
Dev Mehta	2026-06-05
Anaya Desai	2026-06-07
Vivaan Joshi	2026-06-10
Isha Patel	2026-06-12
Krish Shah	2026-06-15
Meera Trivedi	2026-06-18
Yash Thakkar	2026-06-20
Diya Patel	2026-06-22
Arjun Shah	2026-06-25
Kavya Mehta	2026-06-27
Manav Desai	2026-07-01
Pooja Joshi	2026-07-03
Dhruv Patel	2026-07-05

8.SUBQUERIES:-

- GREATER THAN AVG BILL
SELECT *
FROM Patients
WHERE patient_id IN
(
SELECT patient_id
FROM Billing
WHERE amount > (SELECT AVG(amount) FROM Billing)
);

Extra options

	patient_id	name	dob	gender	phone_number	email	address	registration_date
☐	1	Aarav Patel	2001-05-12	Male	9876543210	aarav@gmail.com	Ahmedabad	2026-01-05
☐	3	Dev Mehta	1995-02-15	Male	9876543212	dev@gmail.com	Vadodara	2025-01-15
☐	7	Krish Shah	1992-09-30	Male	9876543216	krish@gmail.com	Surat	2026-02-20
☐	8	Meera Trivedi	1985-12-05	Female	9876543217	meera@gmail.com	Vadodara	2026-03-01
☐	9	Yash Thakkar	1998-07-14	Male	9876543218	yash@gmail.com	Ahmedabad	2026-03-10
☐	11	Arjun Shah	1990-10-11	Male	9876543220	arjun@gmail.com	Ahmedabad	2026-04-01
☐	15	Dhruv Patel	2002-02-27	Male	9876543224	dhruv@gmail.com	Surat	2026-05-10

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

- GREATER THAN AVG FEE
SELECT *
FROM Doctors
WHERE consultation_fee >
(
SELECT AVG(consultation_fee)
FROM Doctors
);

Extra options

	doctor_id	name	specialization	phone_number	email	available_days	consultation_fee
☐	1	Dr. Raj Sharma	Cardiology	9000000001	raj@hospital.com	Monday, Wednesday, Friday	1500.00
☐	3	Dr. Amit Shah	Neurology	9000000003	amit@hospital.com	Monday, Thursday	1800.00
☐	5	Dr. Karan Desai	Orthopedics	9000000005	karan@hospital.com	Tuesday, Friday	1600.00
☐	7	Dr. Vivek Trivedi	Cardiology	9000000007	vivek@hospital.com	Wednesday, Saturday	1700.00

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

EXAM

- APPOINTMENTS WITH DERMATOLOGY DOCTOR

SELECT *

FROM Patients

WHERE patient_id IN

(

SELECT patient_id

FROM Appointments

WHERE doctor_id IN

(

SELECT doctor_id

FROM Doctors

WHERE specialization = 'Dermatology'

)

);

	patient_id	name	dob	gender	phone_number	email	address	registration_date
Edit Copy Delete	2	Riya Shah	1999-08-20	Female	9876543211	riya@gmail.com	Surat	2026-01-10
Edit Copy Delete	12	Kavya Mehta	1997-01-20	Female	9876543221	kavya@gmail.com	Surat	2026-04-05
Edit Copy Delete	14	Pooja Joshi	1994-09-09	Female	9876543223	pooja@gmail.com	Ahmedabad	2026-05-01

Check all 0/3 items selected Edit Copy Delete Export

9. DATE&TIME:-

- APPOINTMENT DATE
SELECT *
FROM Appointments
WHERE MONTH(appointment_date) = 6
AND YEAR(appointment_date) = 2026;

	▼ appointment_id	patient_id	doctor_id	appointment_date	status
Copy  Delete	1	1	1	2026-06-01	Completed
Copy  Delete	2	2	2	2026-06-03	Scheduled
Copy  Delete	3	3	3	2026-06-05	Completed
Copy  Delete	4	4	4	2026-06-07	Completed
Copy  Delete	5	5	5	2026-06-10	Cancelled
Copy  Delete	6	6	6	2026-06-12	Completed
Copy  Delete	7	7	1	2026-06-15	Scheduled
Copy  Delete	8	8	7	2026-06-18	Completed
Copy  Delete	9	9	3	2026-06-20	Completed
Copy  Delete	10	10	4	2026-06-22	Scheduled
Copy  Delete	11	11	5	2026-06-25	Completed
Copy  Delete	12	12	2	2026-06-27	Cancelled

EXAM

- CALCULATE THE TOTAL STAY

```
SELECT patient_id,
       name,
       DATEDIFF(CURDATE(), registration_date) AS days_since_registration
FROM Patients;
```

▼	patient_id	name	days_since_registration
Delete	1	Aarav Patel	274
Delete	2	Riya Shah	269
Delete	3	Dev Mehta	264
Delete	4	Anaya Desai	247
Delete	5	Vivaan Joshi	243
Delete	6	Isha Patel	236
Delete	7	Krish Shah	228
Delete	8	Meera Trivedi	219
Delete	9	Yash Thakkar	210
Delete	10	Diya Patel	205
Delete	11	Arjun Shah	188
Delete	12	Kavya Mehta	184
Delete	13	Manav Desai	177
Delete	14	Pooja Joshi	158
Delete	15	Dhruv Patel	149

- DD-MM-YYYY
SELECT record_id,
patient_id,
DATE_FORMAT(treatment_date,'%d-%m-%Y') AS treatment_date
FROM Medical_Records;

		record_id	patient_id	treatment_date
it	 Copy  Delete	1	1	01-06-2026
it	 Copy  Delete	2	2	03-06-2026
it	 Copy  Delete	3	3	05-06-2026
it	 Copy  Delete	4	4	07-06-2026
it	 Copy  Delete	5	6	12-06-2026
it	 Copy  Delete	6	7	15-06-2026
it	 Copy  Delete	7	8	18-06-2026
it	 Copy  Delete	8	9	20-06-2026
it	 Copy  Delete	9	10	22-06-2026
it	 Copy  Delete	10	11	25-06-2026
it	 Copy  Delete	11	13	01-07-2026
it	 Copy  Delete	12	15	05-07-2026

10. STRING MANIPULATION FUNCTION:-

- UPPERCASE
SELECT UPPER(name) AS patient_name
FROM Patients;

Extra options**patient_name**

AARAV PATEL

RIYA SHAH

DEV MEHTA

ANAYA DESAI

VIVAAN JOSHI

ISHA PATEL

KRISH SHAH

MEERA TRIVEDI

YASH THAKKAR

DIYA PATEL

ARJUN SHAH

KAVYA MEHTA

MANAV DESAI

POOJA JOSHI

DHRUV PATEL

EXAM

- LOWERCASE
SELECT LOWER(name) AS patient_name
FROM Patients;

patient_name

aarav patel

riya shah

dev mehta

anaya desai

vivaan joshi

isha patel

krish shah

meera trivedi

yash thakkar

diya patel

arjun shah

kavya mehta

manav desai

pooja joshi

dhruv patel

EXAM

- NOT AVAILABLE
SELECT name,
COALESCE(email,'Not Available') AS email
FROM Patients;

				name	email
☐				Aarav Patel	aarav@gmail.com
☐				Riya Shah	riya@gmail.com
☐				Dev Mehta	dev@gmail.com
☐				Anaya Desai	anaya@gmail.com
☐				Vivaan Joshi	vivaan@gmail.com
☐				Isha Patel	isha@gmail.com
☐				Krish Shah	krish@gmail.com
☐				Meera Trivedi	meera@gmail.com
☐				Yash Thakkar	yash@gmail.com
☐				Diya Patel	diya@gmail.com
☐				Arjun Shah	arjun@gmail.com
☐				Kavya Mehta	kavya@gmail.com
☐				Manav Desai	manav@gmail.com
☐				Pooja Joshi	pooja@gmail.com
☐				Dhruv Patel	dhruv@gmail.com

11. IMPLIMENT WINDOW FUNCTION:-

- RANK

```

SELECT
    d.doctor_id,
    d.name,
    COUNT(m.record_id) AS patients_treated,
    RANK() OVER (ORDER BY COUNT(m.record_id) DESC) AS doctor_rank
FROM Doctors d
LEFT JOIN Medical_Records m
ON d.doctor_id = m.doctor_id
GROUP BY d.doctor_id, d.name;

```

Extra options

doctor_id	name	patients_treated	doctor_rank
1	Dr. Raj Sharma	3	1
2	Dr. Neha Patel	1	5
3	Dr. Amit Shah	2	2
4	Dr. Priya Mehta	2	2
5	Dr. Karan Desai	1	5
6	Dr. Rina Joshi	2	2
7	Dr. Vivek Trivedi	1	5
8	Dr. Simran Shah	0	8

- CUMULATIVE RENVENUE
SELECT
DATE_FORMAT(payment_date, '%Y-%m') AS month,
SUM(amount) AS monthly_revenue,
SUM(SUM(amount)) OVER (
ORDER BY DATE_FORMAT(payment_date, '%Y-%m')
) AS cumulative_revenue
FROM Billing
WHERE payment_status = 'Paid'
GROUP BY DATE_FORMAT(payment_date, '%Y-%m');

month	monthly_revenue	cumulative_revenue
2026-06	16900.00	16900.00
2026-07	3700.00	20600.00

☐ Show all | Number of rows: 25 ▼ | Filter rows:

- RUNNING TOTAL OF APPOINTMENTS MADE'
SELECT
appointment_id,
patient_id,
doctor_id,
appointment_date,
RANK() OVER (
ORDER BY appointment_date
) AS appointment_rank
FROM Appointments;

EXAM

▼	appointment_id	patient_id	doctor_id	appointment_date	appointment_rank
Delete	1	1	1	2026-06-01	1
Delete	2	2	2	2026-06-03	2
Delete	3	3	3	2026-06-05	3
Delete	4	4	4	2026-06-07	4
Delete	5	5	5	2026-06-10	5
Delete	6	6	6	2026-06-12	6
Delete	7	7	1	2026-06-15	7
Delete	8	8	7	2026-06-18	8
Delete	9	9	3	2026-06-20	9
Delete	10	10	4	2026-06-22	10
Delete	11	11	5	2026-06-25	11
Delete	12	12	2	2026-06-27	12
Delete	13	13	6	2026-07-01	13
Delete	14	14	8	2026-07-03	14
Delete	15	15	1	2026-07-05	15

12.CASE EXPRESSIONS:-

- ASSIGN PAITENT
SELECT
p.patient_id,
p.name,
COUNT(m.record_id) AS total_records,
CASE
WHEN COUNT(m.record_id) > 5 THEN 'High'
WHEN COUNT(m.record_id) = 5 THEN 'Medium'
ELSE 'Low'
END AS risk_level
FROM Patients p
LEFT JOIN Medical_Records m
ON p.patient_id = m.patient_id
GROUP BY p.patient_id, p.name;

Extra options

patient_id	name	total_records	risk_level
1	Aarav Patel	1	Low
2	Riya Shah	1	Low
3	Dev Mehta	1	Low
4	Anaya Desai	1	Low
5	Vivaan Joshi	0	Low
6	Isha Patel	1	Low
7	Krish Shah	1	Low
8	Meera Trivedi	1	Low
9	Yash Thakkar	1	Low
10	Diya Patel	1	Low
11	Arjun Shah	1	Low
12	Kavya Mehta	0	Low
13	Manav Desai	1	Low
14	Pooja Joshi	0	Low
15	Dhruv Patel	1	Low

- CATEGORIZE DOCTORE

ADDING A COLUMN:

ALTER TABLE Doctors

ADD experience_years INT;

UPDATE Doctors SET experience_years = 18 WHERE doctor_id = 1;

UPDATE Doctors SET experience_years = 12 WHERE doctor_id = 2;

UPDATE Doctors SET experience_years = 20 WHERE doctor_id = 3;

UPDATE Doctors SET experience_years = 7 WHERE doctor_id = 4;

UPDATE Doctors SET experience_years = 15 WHERE doctor_id = 5;

UPDATE Doctors SET experience_years = 4 WHERE doctor_id = 6;

UPDATE Doctors SET experience_years = 17 WHERE doctor_id = 7;

UPDATE Doctors SET experience_years = 9 WHERE doctor_id = 8;

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```
ALTER TABLE Doctors ADD experience_years INT;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 1 row affected. (Query took 0.0006 seconds.)

```
UPDATE Doctors SET experience_years = 18 WHERE doctor_id = 1;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 1 row affected. (Query took 0.0006 seconds.)

```
UPDATE Doctors SET experience_years = 12 WHERE doctor_id = 2;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 1 row affected. (Query took 0.0006 seconds.)

```
UPDATE Doctors SET experience_years = 20 WHERE doctor_id = 3;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 1 row affected. (Query took 0.0005 seconds.)

```
UPDATE Doctors SET experience_years = 7 WHERE doctor_id = 4;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 1 row affected. (Query took 0.0003 seconds.)

```
UPDATE Doctors SET experience_years = 15 WHERE doctor_id = 5;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 1 row affected. (Query took 0.0006 seconds.)

```
UPDATE Doctors SET experience_years = 4 WHERE doctor_id = 6;
```

EXAM

```
SELECT
  doctor_id,
  name,
  experience_years,
  CASE
    WHEN experience_years > 15 THEN 'Senior'
    WHEN experience_years BETWEEN 5 AND 15 THEN 'Mid-Level'
    ELSE 'Junior'
  END AS doctor_category
FROM Doctors;
```

	doctor_id	name	experience_years	doctor_category
 Delete	1	Dr. Raj Sharma	18	Senior
 Delete	2	Dr. Neha Patel	12	Mid-Level
 Delete	3	Dr. Amit Shah	20	Senior
 Delete	4	Dr. Priya Mehta	7	Mid-Level
 Delete	5	Dr. Karan Desai	15	Mid-Level
 Delete	6	Dr. Rina Joshi	4	Junior
 Delete	7	Dr. Vivek Trivedi	17	Senior
 Delete	8	Dr. Simran Shah	9	Mid-Level

With selected:  Edit  Copy  Delete  Export

THANK YOU